

ClaimGuard

AI-Powered Insurance Fraud Detection for the Kenyan Motor Market

Shantelle Wambui Kungu | PLP AI/ML Capstone | September 2026

"The technology works. The numbers justify it. Today I'm asking for the go-ahead."

The Problem

KES 40-60 Billion Lost to Motor Insurance Fraud Every Year

5-15 days

Manual review takes 5-15 days per claim -- adjusters catch only 40% of fraud

60% missed

Organised fraud rings file across multiple policyholders -- invisible to a single reviewer

KES 14.4M

A mid-size insurer loses approximately KES 14.4M in undetected fraud every month

The problem isn't that fraud happens. It's that 60% of it goes undetected.

Cost of Inaction

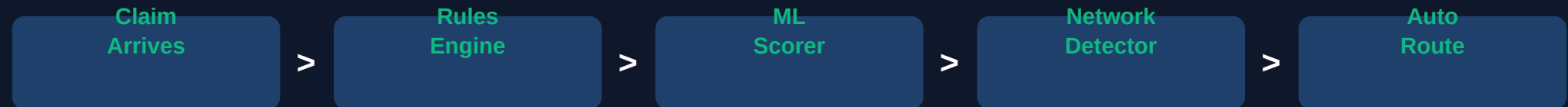
What This Costs Every Month -- Without ClaimGuard

Item	Calculation	Monthly Loss
Undetected fraud	160 claims x KES 150K x 60% missed	KES 14.4M
Adjuster false-positive rework	40% wrongly flagged x rework cost	KES 1.6M
TOTAL VISIBLE MONTHLY LOSS		~KES 16M

Every month we delay costs the business approximately KES 16 million.

Our Solution

ClaimGuard: Three Layers of Intelligence



Rules Engine

8 hard fraud rules:

- * Claim within 14 days of inception
- * Duplicate OB numbers
- * Provider billing spikes
- * Round-sum amounts, missing docs

XGBoost ML Scorer

25 engineered features

Fraud probability: 0 to 1

AUC-ROC: 0.84

Precision: 79% Recall: 81%

Network / Ring Detector

Graph analysis across providers, claimants, and vehicles.

Detects organised rings invisible to human reviewers.

Results

What ClaimGuard Delivers

Metric	Manual	ClaimGuard	Improvement
Processing Time	5-15 days	< 30 seconds	99.7% faster
Fraud Detection Rate	40%	81%	+41 percentage points
Fraud Rings Found	0 (invisible)	12 rings	KES 4.2M surfaced
Manual Review Workload	100%	5% (oversight)	95% reduction

The Business Case

The Numbers That Matter to This Room

Annual Savings

Additional fraud caught KES 118.1M

Adjuster time savings KES 10.9M

False positive reduction KES 19.2M

TOTAL ANNUAL SAVINGS KES 148.2M

Year 1 Investment

Development & integration KES 8.5M

Cloud infrastructure KES 1.8M

Maintenance & retraining KES 1.2M

TOTAL INVESTMENT KES 11.5M

1,188%

Year 1 ROI

28 Days

Payback Period

KES 580M

5-Year NPV

Risks & Mitigation

Risk 1: Regulatory Challenge

IRA or a policyholder challenges an automated rejection

Mitigation

Every rejection includes a full SHAP audit trail. Human review is mandatory for all rejections. Designed to comply with IRA data governance guidelines.

Risk 2: Model Drift

Fraud patterns evolve faster than the model retrain

Mitigation

Automated monthly retraining with champion/challenger model registry. Alert fires if AUC-ROC drops below 0.78, triggering immediate retraining.

The Technology Is Ready

This Is Not a Proposal. It's a Deployed System.

V Backend

Python/FastAPI -- production-grade, containerised with Docker

V Database

PostgreSQL -- 22-table schema, migrations managed with Alembic

V ML Pipeline

XGBoost trainer, SHAP explainer, model registry with promote/rollback

V Frontend

Next.js -- case review, HITL queue, network visualisation, ML admin

V Security

JWT auth, 8 user roles, full audit trail on every single decision

V Scalability

Celery + Redis async processing, horizontal scaling ready

What We're Asking For

Three Approvals. One Decision.

1

Budget: KES 11.5M (Year 1)

Development & integration sprint

Cloud infrastructure

Compliance review

2

IT Window: Q4 2026

90-day integration with InsureMaster

Zero disruption to existing systems

3

Compliance Sign-off

IRA data governance review

Full documentation provided

Go. Deploy ClaimGuard.

KES 11.5M investment.

KES 148M return in Year 1.

Payback in 28 days.

The technology works.

Questions?